

Minor-1 Notes

UNIT-I

BJT → Introduction & types of transistors
Construction, BJT Characteristics
in CB, CE & CC mode, operating
point, AC/DC load line, leakage
currents, Saturation and Cut-off
mode of operations, Ebers Moll
Model.

Bias Stabilization → Need of stabilization,
Various biasing schemes, bias
stability w.r.t. variation in I_{CO} , V_{BE} , β

Models → Stabilization factor, thermal stability
low frequency models for transistors
(h-parameter, Hybrid π , δ_e)

UNIT-II

BJT Amplifiers → Analysis of low freq. (CB, CE,
CC & CC with R_E), Comparison
of diff configurations.

⇒ Assignment Topics → Cascaded Amp., Darlington Pair,
Cascode Amplifiers.

1904-1947 - vacuum tube electronics
 1904 - J.A. Fleming.

Dec. 23, 1947 - Transistor - by 1. Dr. William Shockley
 2. Dr. Walter H. Brattain
 3. Dr. John Bardeen.

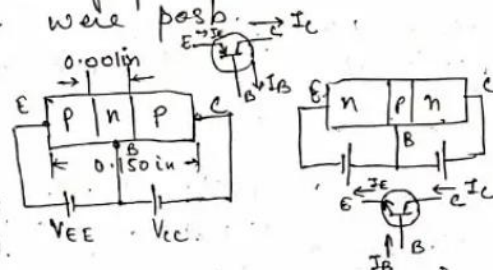
✓ 3. term. solid state device.

Adv. ^{over} Vacuum tube -

1. was smaller & light-wt.
2. had no heater requirement or heater loss.
3. had rugged construction
4. more efficient since less power absorbed by device itself.
5. was instantly available for use.
6. requiring no warm-up period
7. lower op'g volt. were poss.

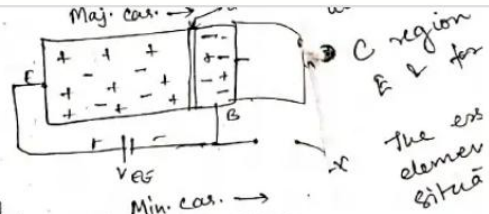
Transistor Construction →

- 3 layer SC device.
- consists of either - 2n & 1p-type (n-p-n) or 2p & 1n-type (p-n-p).
- dc biasing - necessary to establish proper region of op'n for ac amplification.
- E-layer: heavily doped, B-layer: lightly doped
 E-layer: lightly (but more than B) doped.
- width: $w_C > w_E > w_B$.
- Outer layer width & inner sandwiched layer $w = 100:1$
- Outer layer dopin & sandwiched layer d. = 10:1 (or less)
- purpose of low doping - conductivity decreases.
- Term 'Bipolar' - holes & e's both participate in injection process into oppositely polarized material.

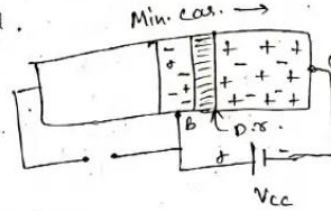


(1)

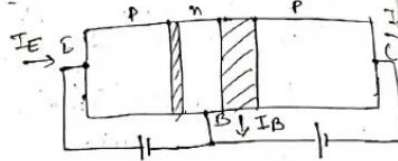
- forward-biased diode.
- d.r. has been reduced in width coz of biasing.
- resulting heavy flow of maj. car. from p $\rightarrow$ n type material.



- reverse-biased diode
- width of d.r. $\uparrow$ d.
- flow of maj. car. is nearly zero & only min. car. flow.



- large no. of maj. car. will diffuse across f.b. junction in n-type mate.



- Since sandwiched n-type mate. (B) is v. thin & lightly doped (low conductⁿ), $\rightarrow$ a v. small no. of these car. (from E $\rightarrow$ B), will take this path of high resis. to B terminal (to create I_B). $I_B \rightarrow \mu A$, I_C & $I_E \rightarrow mA$.

- A large no. of these maj. car. (E $\rightarrow$ B) will diffuse across r.b. junction into p-type (C)

How? - Reason for relative ease $\leftarrow$ maj. car. can cross r.b. diode junction, if we consider that for r.b. diode injected maj. car. will appear as min. car. in n-type mate. (B).

- Applying KCL to transistor - as if it were a single node $\rightarrow$

$$I_E = I_C + I_B \quad \text{--- (1)}$$